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Roll No. :.....

CSA-101
Examination – June - 2023
B.Tech. I Sem : CSE, IT, AI, BC, IoT
Introduction to Computer Science Engineering

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Which of the following has the fastest speed in the computer memory hierarchy? 02

A. Cache B. Register in CPU C. Main memory D. Disk cache

(b) Fill in the blank: 02

_____ keeps track of the address of the next instruction.

(c) What are the steps taken by the CPU to execute the instructions? 03

OR

What is a bus? Describe the functions of a control bus. 03

(d) What is a ROM? Differentiate between PROM, EPROM and EEPROM. 05

OR

What is software? What are its different types? Explain each in detail. 05

Q.2 (a) Relational operators cannot be used on _____ 02

A. structure B. long C. strings D. float

(b) Fill in the blank: 02

In Implicit type conversion, If an operand of long int is present then the other operand will be converted to .....

(c) Explain the structure of C program. 03

OR

What are data types? Explain its types and its type modifiers. 03

(d) Write a C program to find out whether it is an odd number or even number. 05

OR

Explain the following control constructs- 05
If-else, for, while and do-while.

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Q.3 (a) What is an Array in C language? 02

A. A group of elements of same data type.

B. An array contains more than one element

C. Array elements are stored in memory in continuous or contiguous locations.

D. All the above.

(b) Prior to using a pointer variable it should be 02

A. Declared B. Initialized C. Both declared and initialized D. None of these

(c) What do you mean by scope of a variable? 03

OR

Differentiate call by value and call by reference. 03

(d) Write a C program to find the average marks obtained by a class of 10 students in a test. 05

OR

What is recursion? Write a simple C program to show the application of recursion. 05

Q.4 (a) A user defined data type, which is used to assign names to integral constants is called 02

_____

(b) What is the default return type if it is not specified in function definition? 02

A. void B. int C. double D. short int

(c) How is array of structure created? 03

OR

What is pointer to structure? 03

(d) What are preprocessor directives? Also explain at least two with example. 05

OR

What is the use of structure in C programming language? 05

Q.5 (a) Which of the following is not a type of cloud server? 02

A. Public Cloud Servers B. Private Cloud Servers

C. Dedicated Cloud Servers D. Merged Cloud Servers

(b) What is a blockchain? 02

A. Currency B. centralized ledger

C. type of cryptocurrency D. distributed ledger on a peer to peer network

(c) What is Web Development? 03

OR

What is the use of an operating system? 03

(d) What is Data Science? Explain the lifecycle of Data Science. 05

OR

How Machine Learning works? Also give the classification of Machine Learning. 05

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